

Problem

If the load impedance is $20 - j20$, the power factor is

- (a) $\angle -45^\circ$
- (b) 0
- (c) 1
- (d) 0.7071
- (e) none of these

Step-by-step solution

Step 1 of 3

Consider the load impedance is $Z_L = 20 - j20$.

Convert the load impedance into polar form.

$$Z_L = 20 - j20$$

Consider the general representation of the complex form.

$$Z_L = a + jb$$

Determine the magnitude.

$$\begin{aligned} |Z_L| &= \sqrt{a^2 + b^2} \\ |Z_L| &= \sqrt{20^2 + 20^2} \\ &= \sqrt{800} \\ &= 28.28 \end{aligned}$$

Step 2 of 3

Determine the phase angle.

$$\begin{aligned} \theta &= \tan^{-1}\left(\frac{b}{a}\right) \\ \theta &= \tan^{-1}\left(-\frac{20}{20}\right) \\ &= \tan^{-1}(-1) \\ &= -45^\circ \end{aligned}$$

Step 3 of 3

Determine the power factor.

Consider the equation for power factor.

$$\begin{aligned}\text{power factor} &= \cos(\theta) \\ &= \cos(-45^\circ) \\ &= 0.7071\end{aligned}$$

Therefore, the power factor is 0.7071.

Therefore, the correct option is d.